

Spring 1997

Problem 1 (Su80, Sp97, Fa11, Fa13, Fa17, Sp21, Fa25). For each $(a, b, c) \in \mathbb{R}^3$, consider the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b \log^c n}$$

Determine the values of (a, b, c) for which the series converges absolutely; converges but not absolutely; diverges.

Problem 2 (Sp97). Let M be a metric space with metric d . Let C be a nonempty closed subset of M . Define $f : M \rightarrow \mathbb{R}$ by

$$f(x) = \inf\{d(x, y) \mid y \in C\}$$

Show that f is continuous, and that $f(x) = 0$ if and only if $x \in C$.

Problem 3 (Fa90, Sp97). Let f be a continuous real valued function satisfying $f(x) \geq 0$, for all x , and

$$\int_0^{\infty} f(x) dx < \infty$$

Prove that

$$\frac{1}{n} \int_0^n x f(x) dx \rightarrow 0$$

as $n \rightarrow \infty$.

Problem 4 (Sp97). Let f and g be two entire functions such that, for all $z \in \mathbb{C}$, $\Re f(z) \leq k \Re g(z)$ for some real constant k . Show that there are constants a, b such that $f(z) = ag(z) + b$.

Problem 5 (Sp97). Prove that $\int_{-\infty}^{\infty} \frac{e^{-(t-i\gamma)^2/2}}{\sqrt{2\pi}} dt$ is independent of the real parameter γ .

Problem 6 (Sp97). Suppose that X is a topological space and V is a finite dimensional subspace of the vector space of continuous real valued functions on X . Prove that there exist a basis $\{f_1, \dots, f_n\}$ for V and points $x_1, \dots, x_n$ in X such that $f_i(x_j) = \delta_{ij}$.

Problem 7 (Fa85, Sp97, Fa98). Suppose that A and B are endomorphisms of a finite dimensional vector space V over a field $\mathbb{F}$. Prove or disprove the following statements:

1. Every eigenvector of AB is also an eigenvector of BA .
2. Every eigenvalue of AB is also an eigenvalue of BA .

Problem 8 (Sp79, Sp97, Sp24). Classify all abelian groups of order 80 up to isomorphism.

Problem 9 (Sp97). Let R be the ring of $n \times n$ matrices over a field. Suppose S is a ring and $h : R \rightarrow S$ is a homomorphism. Show that h is either injective or zero.

Problem 10 (Fa88, Sp97). Let f be a function from $[0, 1]$ into itself whose graph

$$G_f = \{(x, f(x)) \mid x \in [0, 1]\}$$

is a closed subset of the unit square. Prove that f is continuous.
Note: See also ????

Problem 11 (Sp97). Suppose that $f''(x) = (x^2 - 1)f(x)$ for all $x \in \mathbb{R}$, and that $f(0) = 1$ and $f'(0) = 0$. Show that $f(x) \rightarrow 0$ as $x \rightarrow \infty$.

Problem 12 (Sp78, Sp83, Sp97). Evaluate $\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx$.

Problem 13 (Fa88, Sp97). Determine the group $\text{Aut}_{\mathbb{C}}$ of all one-to-one analytic maps of $\mathbb{C}$ onto $\mathbb{C}$.

Problem 14 (Sp97, Sp17). Show that

$$\det e^M = e^{\text{tr } M}$$

for any complex $n \times n$ matrix M , where e^M is defined as in ??.

Problem 15 (Sp97). Suppose that P and Q are $n \times n$ matrices such that $P^2 = P$, $Q^2 = Q$, and $1 - (P + Q)$ is invertible. Show that P and Q have the same rank.

Problem 16 (Sp97). Suppose that A is a commutative algebra with identity over $\mathbb{C}$, i.e., A is a commutative ring containing $\mathbb{C}$ as a subring with identity. Suppose further that $a^2 \neq 0$ for all nonzero elements $a \in A$. Show that if the dimension of A as a vector space over $\mathbb{C}$ is finite and at least two, then the equations $a^2 = a$ is satisfied by at least three distinct elements $a \in A$.

Problem 17 (Sp97). Let $\text{GL}_2(\mathbb{Z}_m)$ denote the multiplicative group of invertible 2×2 matrices over the ring of integers modulo m . Find the order of $\text{GL}_2(\mathbb{Z}_{p^n})$ for each prime p and positive integer n .

Problem 18 (Sp97). Let H be the quotient of an abelian group G by a subgroup K . Prove or disprove each of the following statements:

1. If H is finite cyclic then G is isomorphic to the direct product of H and K .
2. If H is a direct product of infinite cyclic groups then G is isomorphic to the direct product of H and K .